Tamanjyot Singh

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Education

The University of Texas at Dallas — Richardson, TX, USA
Doctorate in Physics — *Expected May 2026*

Thesis: The Black Hole Ballet: Spin and precession effects on gravitational waves and tidal disruption Events

Advisor: Dr. Michael Kesden

The University of Texas at Dallas — Richardson, TX, USA
Master of Science in Physics, GPA: 3.93/4.0 — *December 2023*

Punjab Engineering College (Deemed to be University) — Chandigarh, India
Bachelor of Technology in Aerospace Engineering with Minor Specialization in Mathematics, GPA : 9.2/10.0, Rank 2 — *June 2020*

Major project: Computational study of flame propagation in a micro-scale channel and applications of such micro-combustors in thermo-electric power generation

Experience

The University of Texas at Dallas — TX, USA
Research Assistant — *May 2022 - Present*

Supervisor: Dr. Michael Kesden

- **Tidal disruption events**: Solving for inclination biases induced by spin breaking the spherical symmetry in stellar disruptions by supermassive black holes.
  Meeting weekly, developing a code, supervising undergraduate students, preparing, and submitting a manuscript resulting in a published research article in Physical Reviews.
- **Precession in gravitational waves**: Investigating the effects of spin dynamics in a binary black hole system on the gravitational wave emission.
  Meeting weekly, developing a code, working in a team environment, supervising undergraduate students, and preparing a manuscript to be submitted for publication.
- **Lensing and Precession**: Distinguishing between a lensed and precessing gravitational wave signal, exploring degeneracies and establishing differences.
  Meeting weekly, developing a code, supervising a post-baccalaureate student, and preparing a manuscript to be submitted for publication.

The University of Texas at Dallas — TX, USA
Teaching Assistant — *January 2021 - May 2022*

- **Physics Laboratory I: Mechanics**: Hosting weekly sessions for students to go through the Mechanics Lab experiments. Grading their Lab report submissions

Indian Institute of Indore — Indore, India
Masters of Science in Astronomy — *August 2020 - December 2020*

Dropped out: Attended 1 semester online (Courses: Astrophysical fluids and plasma, Relativity and cosmology, etc.)

Combustion research Laboratory, IIT Bombay — Mumbai, India
Research Intern — *January 2019 - April 2019*

- **Combustion characteristics and the effect of hydrogen content on syngas-air mixtures**: Formulated an Analytical correlation for Laminar Burning Velocity of Multi-component Gaseous Mixtures Simultaneously at Elevated Pressures and Temperatures.

3 BRD, Air Force Station — Chandigarh, India
IAF (Indian Air Force) Intern — *June 2018*

Publications

Submitted papers: ...

Published papers:

Singh, T. and M. Kesden (2024, February) : "Distribution of orbital inclinations for tidal disruption events by Kerr black holes." 109 (4), 043016

Talks

- Spring meeting 2023, Texas section APS, Texas A&M University, Commerce, TX — March 2023
  Contributed Talk: Tidal disruption events by spinning supermassive black holes
- Fall meeting 2023, Texas section APS, Angelo State University, San Angelo, TX — October 2023
  Contributed Talk: Identifying Precession in Gravitational Wave signals from Binary Black Hole inspIrals
- International Conference on Gravitation and Cosmology 2023, IIT Guwahati, Assam, India — December 2023
  Contributed Talk: Inclination distributions in tidal disruption events by spinning supermassive black holes
- 10th Gulf Coast Gravity meeting 2024, UT Austin, TX — March 2024
  Contributed Talk: Gravitational wave signals from precessing binary black hole inspirals
- TEXAS PAARE Annual Symposium (UTD/UNT symposium) 2024, UNT, Denton, TX — March 2024
  Contributed Talk: Tidal disruption events: Inclination distributions for spinning supermassive black holes
- APS April meeting 2024, Sacramento, CA — April 2024
  Contributed Talk: Gravitational wave signals from precessing binary black hole inspirals
- Center for Computational Astrophysics, Flatiron Institute, New York city, NY — May 2024
  Invited visit: Precession in GWs

Notable Achievements

- **APS April meeting DGRAV Travel Award** 2024
- **Margie Renfrow travel award** 2024 by the Department of Physics, The University of Texas at Dallas, TX
- **Teaching Assistant award** 2022 at The University of Texas at Dallas, TX
- ***PECOSA Sydney Prize*** for year 2017-18 (Merit Based - Awarded to 3 out of 700+ students)
- Secured **1st position** in the Mathematics Quiz sponsored by Punjab State Council for Science and Technology, Chandigarh (Gold Medal - District Level Competition)
- **99.3%ile** in JEE (Mains) 2016 taken by 1,400,000+ candidates all over India
- **International Rank 76** in IOS (International Olympiad of Science) 2013 (Gold Medal + *Special Achievement Certificate*)

Workshops and Extra Course Certificates

- Course Certificate for **Astro 101 : Black Holes** by University of Alberta on Coursera. (June 2020)
- Hosted a workshop at undergrad level for **Gravitational lensing** at Punjab Engineering College, Chandigarh (October 2019)
- Attended **ASTROWIN** , Winter School on Astronomy (February 2019)
- Course Certificate for **Particle Physics: an Introduction** by University of Geneva on Coursera. (July 2018)
- Course Certificate for **Understanding Einstein: The Special Theory of Relativity** by Stanford University on Coursera. (July 2017)

Skills

- **Programming Languages**: Python, C++, SQL, Matlab
- **Softwares**: pyLIMA, Fluent, AutoCAD, OriginLab, Athena++, Jupyter, Solidworks and ANSYS